In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```
df=pd.read_csv("Ecommerce_Sales_Data_2024_2025 (2).csv")
```

In [3]:
```
df
```

Out[3]:

	Order ID	Order Date	Customer Name	Region	City	Category	Sub-Category	Product Name	Quantity	Un Pric
0	10001	19-10-2024	Kashvi Varty	South	Bangalore	Books	Non-Fiction	Non-Fiction Ipsum	2	3629
1	10002	30-08-2025	Advik Desai	North	Delhi	Groceries	Rice	Rice Nemo	1	4216
2	10003	04-11-2023	Rhea Kalla	East	Patna	Kitchen	Juicer	Juicer Odio	4	6487
3	10004	23-05-2025	Anika Sen	East	Kolkata	Groceries	Oil	Oil Doloribus	5	3732
4	10005	19-01-2025	Akarsh Kaul	West	Pune	Clothing	Kids Wear	Kids Wear Quo	1	5003
...	...	...	...	...	...	...	...	...	...	
4995	14996	25-06-2024	Nishith Kulkarni	East	Kolkata	Books	Fiction	Fiction Veritatis	3	6067
4996	14997	22-12-2024	Aaina Chander	North	Jaipur	Toys	Doll	Doll Nulla	5	7004
4997	14998	15-04-2025	Dhanush Gara	South	Bangalore	Beauty	Lipstick	Lipstick Eaque	1	4216
4998	14999	08-07-2024	Divyansh Malhotra	East	Kolkata	Electronics	Smartwatch	Smartwatch Adipisci	4	1356
4999	15000	04-02-2024	Aarush Walla	West	Goa	Clothing	Kids Wear	Kids Wear Repellat	1	7676

5000 rows × 14 columns

In [4]:
```
df.head()
```

Out[4]:

	Order ID	Order Date	Customer Name	Region	City	Category	Sub-Category	Product Name	Quantity	Unit Price	Disco
0	10001	19-10-2024	Kashvi Varty	South	Bangalore	Books	Non-Fiction	Non-Fiction Ipsum	2	36294	
1	10002	30-08-2025	Advik Desai	North	Delhi	Groceries	Rice	Rice Nemo	1	42165	
2	10003	04-11-2023	Rhea Kalla	East	Patna	Kitchen	Juicer	Juicer Odio	4	64876	
3	10004	23-05-2025	Anika Sen	East	Kolkata	Groceries	Oil	Oil Doloribus	5	37320	
4	10005	19-01-2025	Akarsh Kaul	West	Pune	Clothing	Kids Wear	Kids Wear Quo	1	50037	

In [5]:
```
df.shape
```

Out[5]:
```
(5000, 14)
```

In [6]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 14 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Order ID       5000 non-null   int64
 1   Order Date     5000 non-null   object
 2   Customer Name  5000 non-null   object
 3   Region         5000 non-null   object
 4   City           5000 non-null   object
 5   Category       5000 non-null   object
 6   Sub-Category   5000 non-null   object
 7   Product Name   5000 non-null   object
 8   Quantity       5000 non-null   int64
 9   Unit Price     5000 non-null   int64
 10  Discount       5000 non-null   int64
 11  Sales          5000 non-null   float64
 12  Profit         5000 non-null   float64
 13  Payment Mode   5000 non-null   object
dtypes: float64(2), int64(4), object(8)
memory usage: 547.0+ KB
```

In [7]:
```
df.describe()
```

Out[7]:

	Order ID	Quantity	Unit Price	Discount	Sales	Profit
count	5000.000000	5000.000000	5000.000000	5000.000000	5000.000000	5000.000000
mean	12500.500000	2.992600	39760.904600	10.051000	106733.204870	15941.746982
std	1443.520003	1.413133	22831.783946	7.084662	85108.208202	14897.684916
min	10001.000000	1.000000	222.000000	0.000000	264.100000	19.120000
25%	11250.750000	2.000000	20312.250000	5.000000	39766.537500	4892.295000
50%	12500.500000	3.000000	39459.500000	10.000000	83080.325000	11108.525000
75%	13750.250000	4.000000	59721.750000	15.000000	156968.587500	22467.987500
max	15000.000000	5.000000	79998.000000	20.000000	398485.000000	89688.440000

In [8]:
```
df.isnull().sum()
```

Out[8]:
```
Order ID         0
Order Date       0
Customer Name    0
Region           0
City             0
Category         0
Sub-Category     0
Product Name     0
Quantity         0
Unit Price       0
Discount         0
Sales            0
Profit           0
Payment Mode     0
dtype: int64
```

In [9]:
```
df.duplicated().sum()
```

Out[9]:
```
np.int64(0)
```

In [10]:
```
df["Order Date"] =pd.to_datetime(df["Order Date"])
```

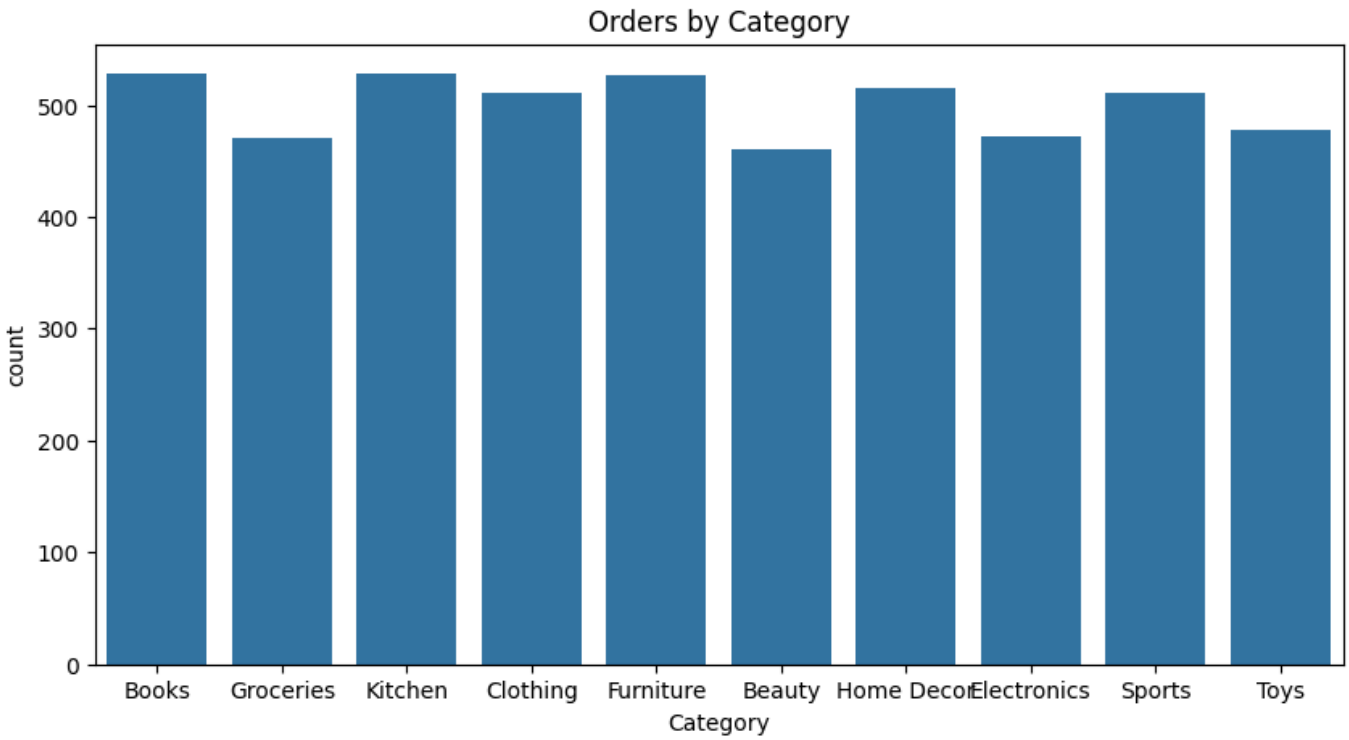
```
C:\Users\HP\AppData\Local\Temp\ipykernel_3316\1104300470.py:1: UserWarning: Parsing dates in %
d-%m-%Y format when dayfirst=False (the default) was specified. Pass `dayfirst=True` or specif
y a format to silence this warning.
  df["Order Date"] =pd.to_datetime(df["Order Date"])
```

In [11]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 14 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Order ID       5000 non-null   int64
 1   Order Date     5000 non-null   datetime64[ns]
 2   Customer Name  5000 non-null   object
 3   Region         5000 non-null   object
 4   City           5000 non-null   object
 5   Category       5000 non-null   object
 6   Sub-Category   5000 non-null   object
 7   Product Name   5000 non-null   object
 8   Quantity       5000 non-null   int64
 9   Unit Price     5000 non-null   int64
 10  Discount       5000 non-null   int64
 11  Sales          5000 non-null   float64
 12  Profit         5000 non-null   float64
 13  Payment Mode   5000 non-null   object
dtypes: datetime64[ns](1), float64(2), int64(4), object(7)
memory usage: 547.0+ KB
```

In [12]:

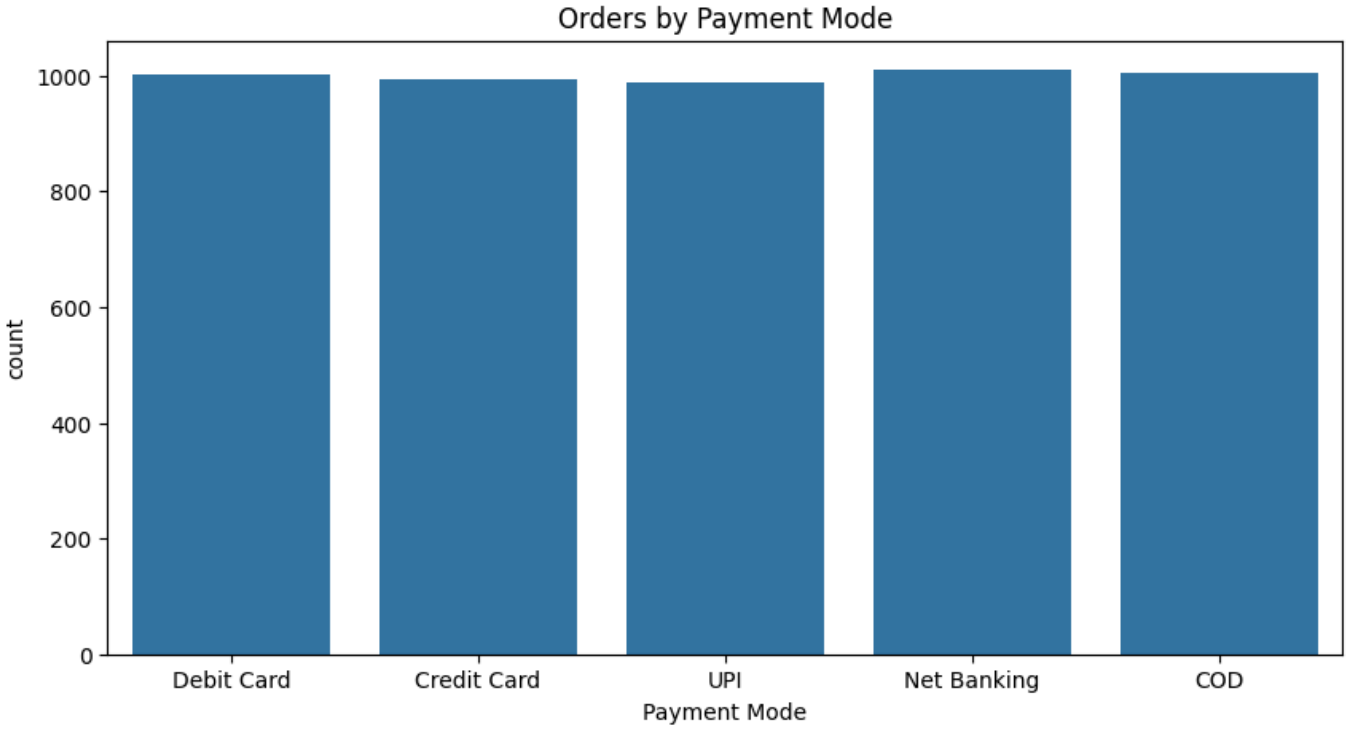
```
plt.figure(figsize=(10,5))
sns.countplot(x="Category", data=df)
plt.title("Orders by Category")
plt.show()
```



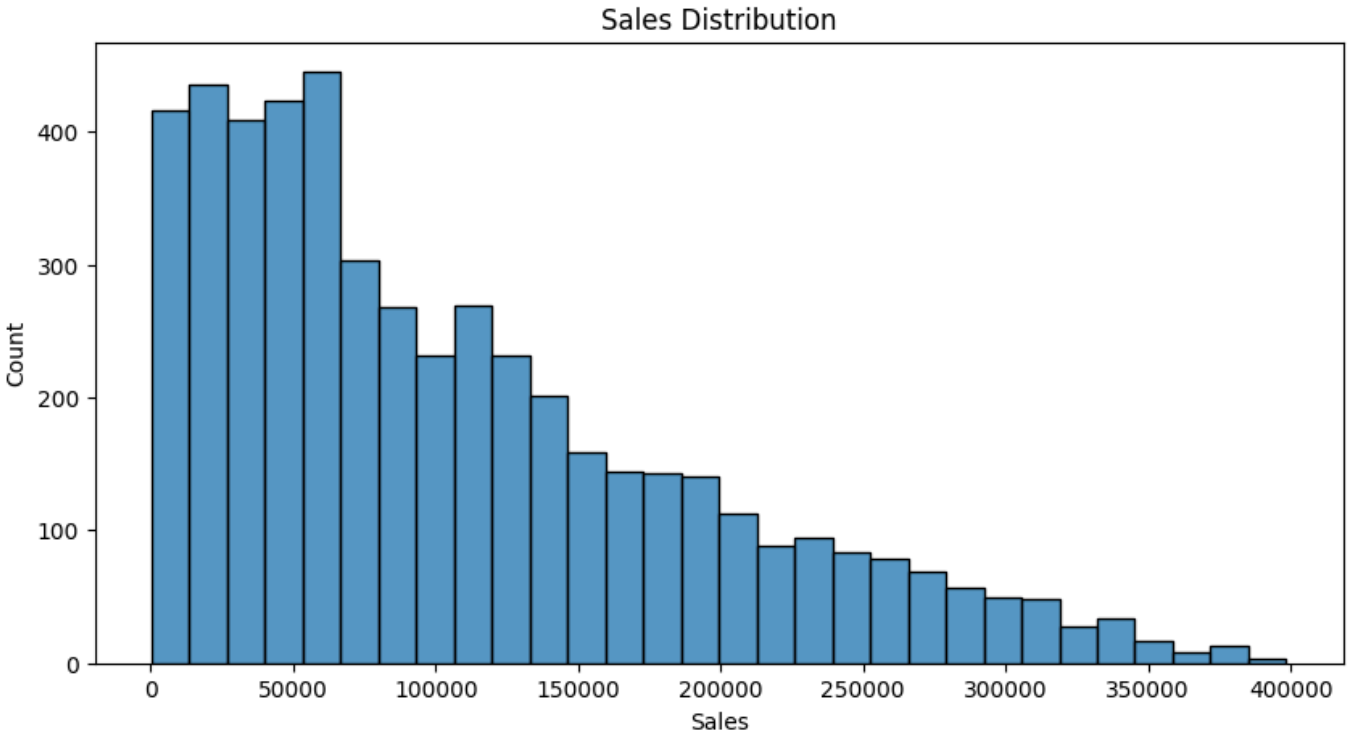
In [13]:

```
plt.figure(figsize=(10,5))
sns.countplot(x="Region", data=df)
plt.title("Orders by Region")
plt.show()
```

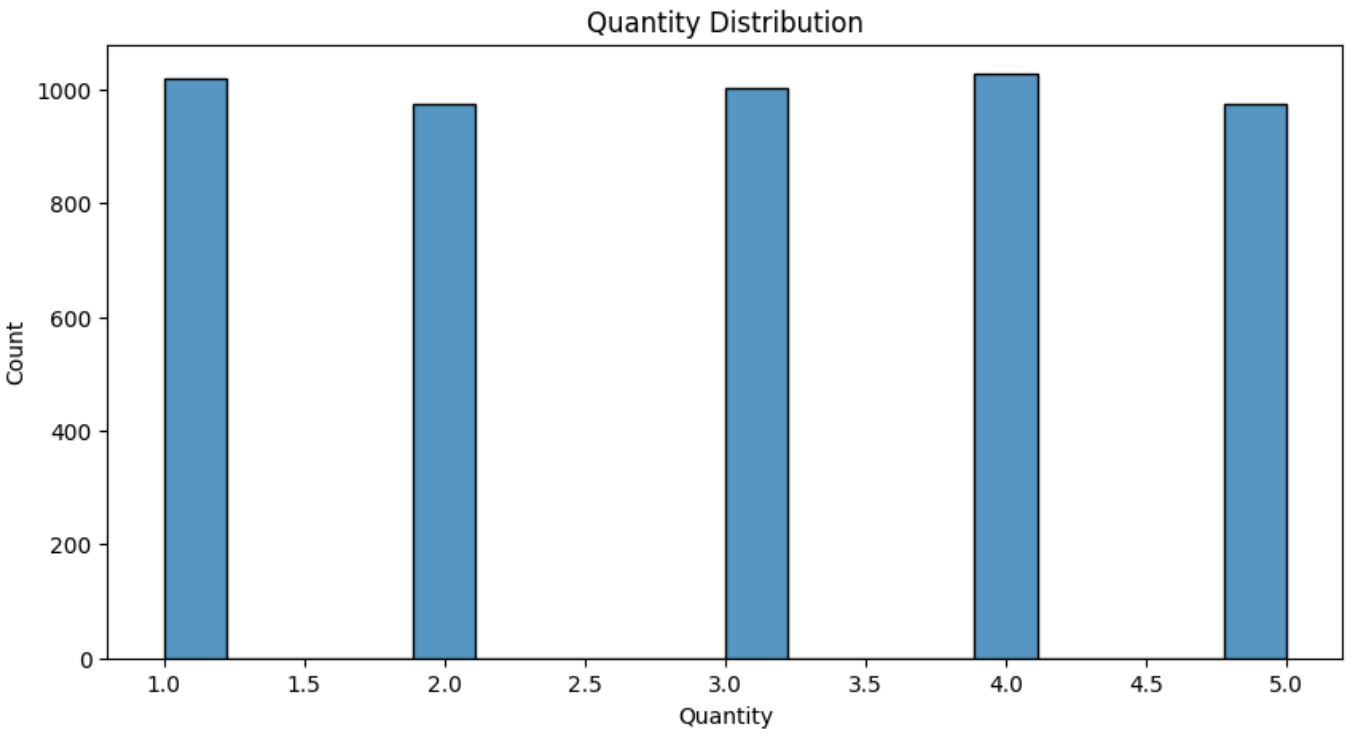
In [14]:
```
plt.figure(figsize=(10,5))
sns.countplot(x="Payment Mode", data=df)
plt.title("Orders by Payment Mode")
plt.show()
```



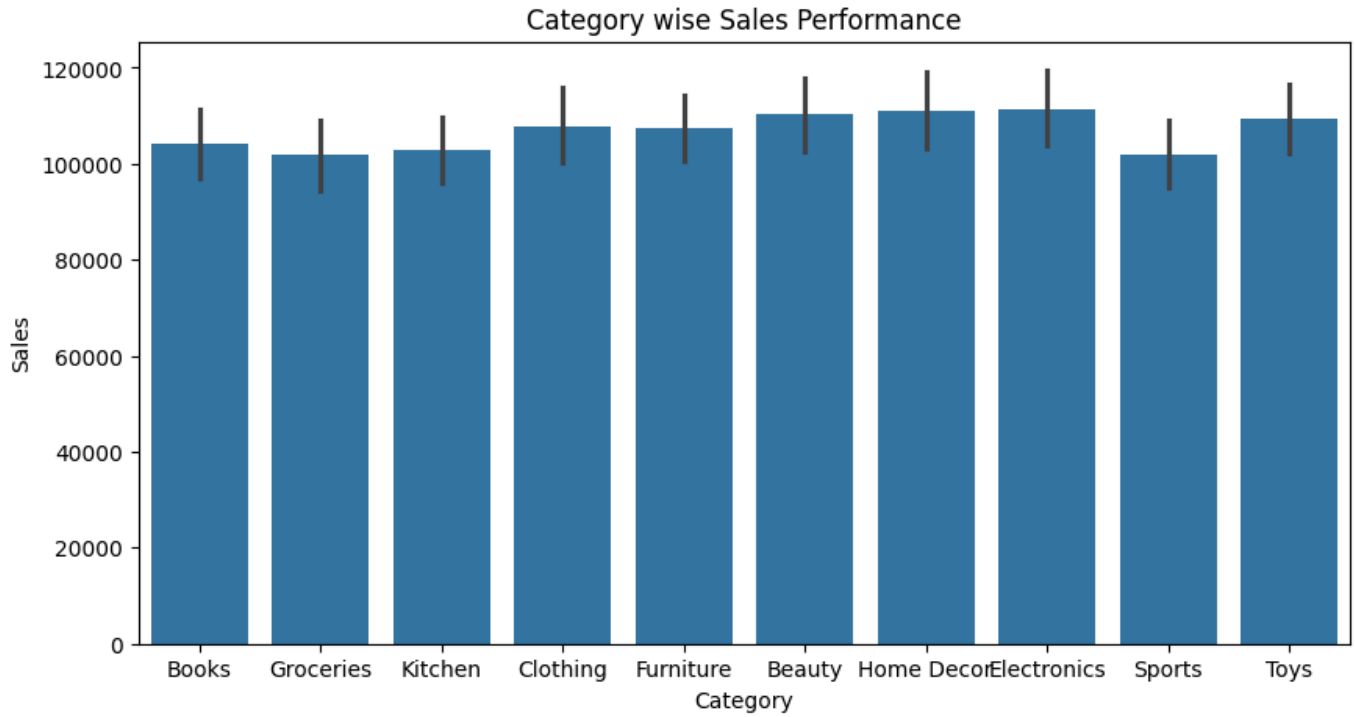
In [15]:
```
plt.figure(figsize=(10,5))
sns.histplot(x="Sales", data=df)
plt.title("Sales Distribution")
plt.xlabel("Sales")
plt.show()
```



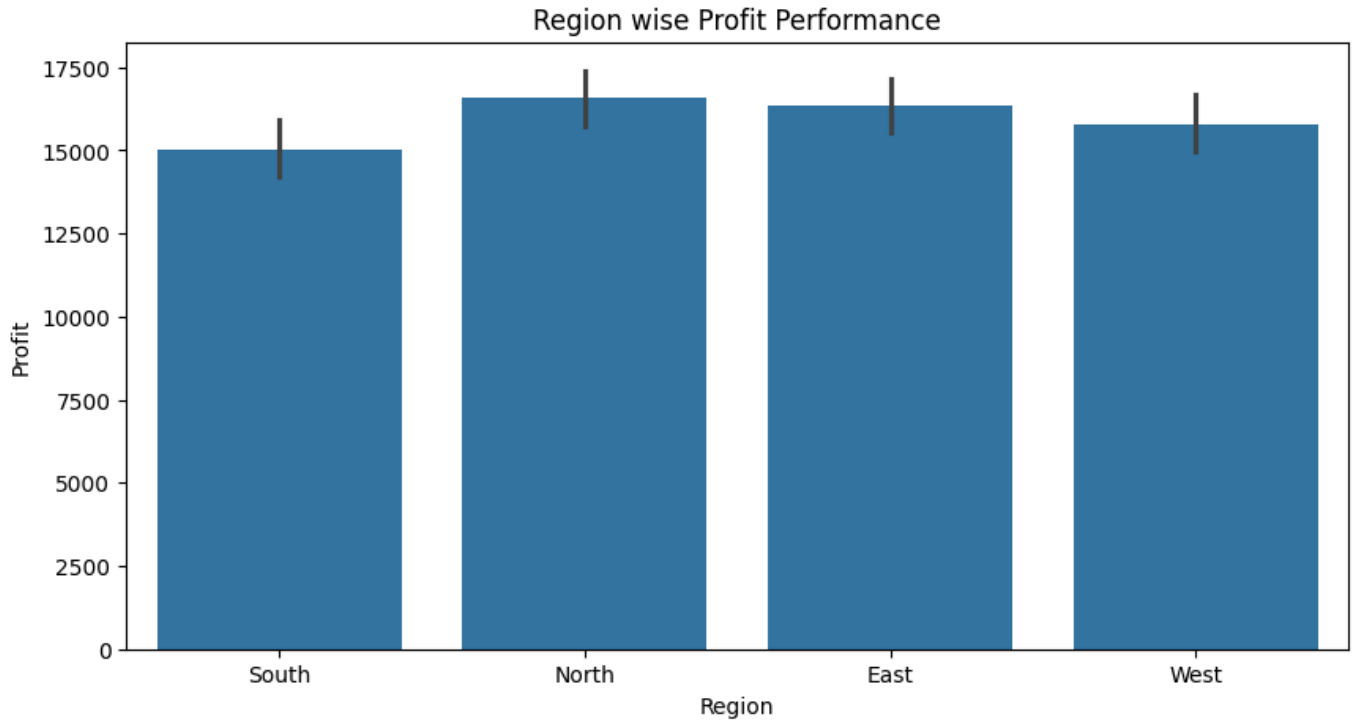
In [16]:
```
plt.figure(figsize=(10,5))
sns.histplot(x="Quantity", data=df)
plt.title("Quantity Distribution")
plt.xlabel("Quantity")
plt.show()
```



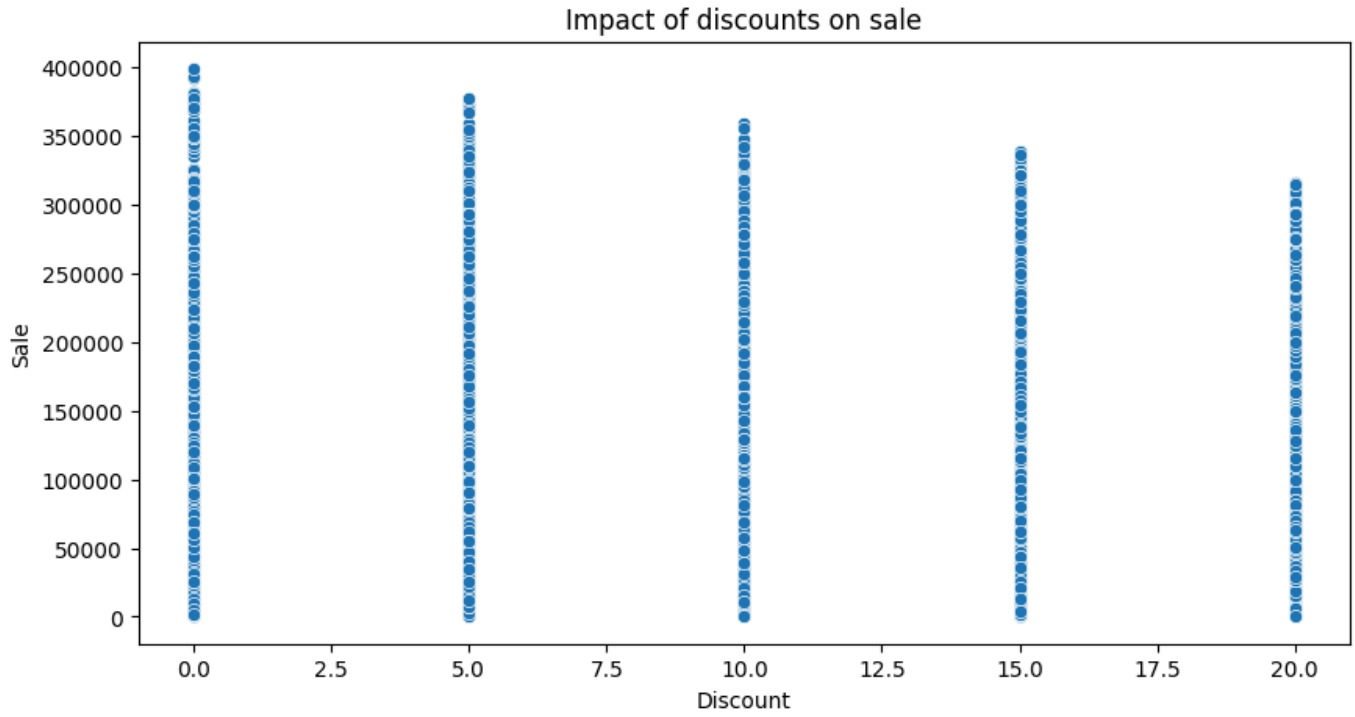
In [18]:
```
plt.figure(figsize=(10,5))
sns.barplot(x="Category", y="Sales",data=df)
plt.title("Category wise Sales Performance")
plt.xlabel("Category")
plt.ylabel("Sales")
plt.show()
```



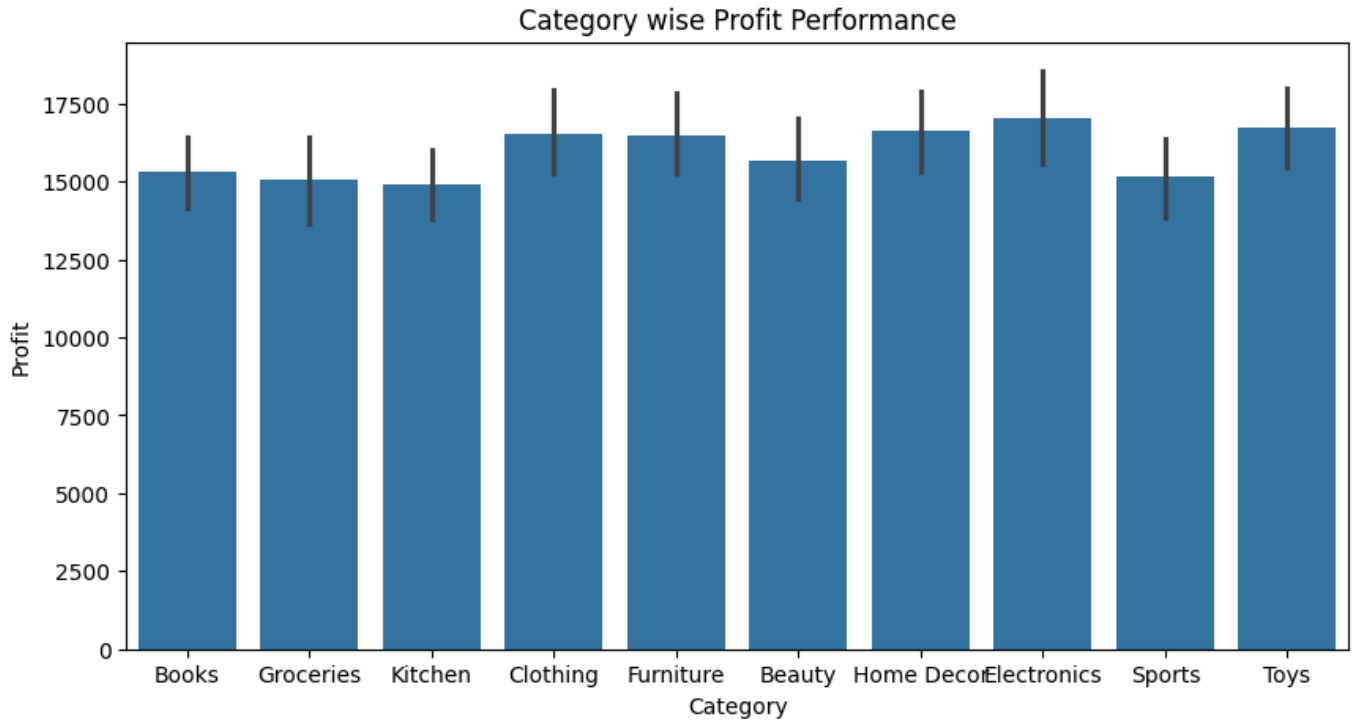
In [19]:
```
plt.figure(figsize=(10,5))
sns.barplot(x="Region", y="Profit",data=df)
plt.title("Region wise Profit Performance")
plt.xlabel("Region")
plt.ylabel("Profit")
plt.show()
```



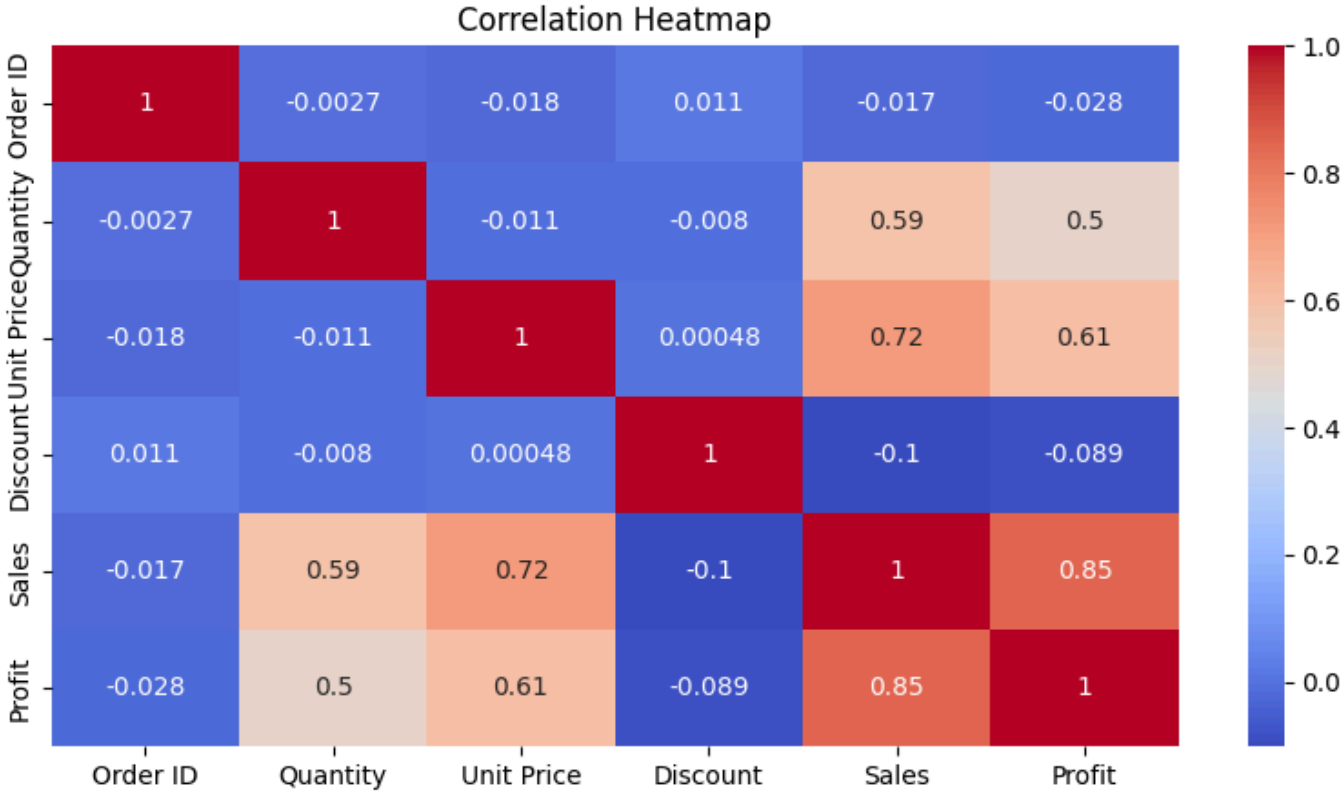
In [21]:
```
plt.figure(figsize=(10,5))
sns.scatterplot(x="Discount", y="Sales",data=df)
plt.title("Impact of discounts on sale")
plt.xlabel("Discount")
plt.ylabel("Sale")
plt.show()
```



In [23]:
```
plt.figure(figsize=(10,5))
sns.barplot(x="Category", y="Profit",data=df)
plt.title("Category wise Profit Performance")
plt.xlabel("Category")
plt.ylabel("Profit")
plt.show()
```



In [26]:
```
plt.figure(figsize=(10,5))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```



In [28]:
```
df["Revenue_per_Item"] = df["Unit Price"] * df["Quantity"]
```

In [29]:
```
df
```

Out[29]:

	Order ID	Order Date	Customer Name	Region	City	Category	Sub-Category	Product Name	Quantity	Un Pric
0	10001	2024-10-19	Kashvi Varty	South	Bangalore	Books	Non-Fiction	Non-Fiction Ipsum	2	3629
1	10002	2025-08-30	Advik Desai	North	Delhi	Groceries	Rice	Rice Nemo	1	4216
2	10003	2023-11-04	Rhea Kalla	East	Patna	Kitchen	Juicer	Juicer Odio	4	6487
3	10004	2025-05-23	Anika Sen	East	Kolkata	Groceries	Oil	Oil Doloribus	5	3732
4	10005	2025-01-19	Akarsh Kaul	West	Pune	Clothing	Kids Wear	Kids Wear Quo	1	5003
...	...	...	...	...	...	...	...	...	...	
4995	14996	2024-06-25	Nishith Kulkarni	East	Kolkata	Books	Fiction	Fiction Veritatis	3	6067
4996	14997	2024-12-22	Aaina Chander	North	Jaipur	Toys	Doll	Doll Nulla	5	7004
4997	14998	2025-04-15	Dhanush Gara	South	Bangalore	Beauty	Lipstick	Lipstick Eaque	1	4216
4998	14999	2024-07-08	Divyansh Malhotra	East	Kolkata	Electronics	Smartwatch	Smartwatch Adipisci	4	1356
4999	15000	2024-02-04	Aarush Walla	West	Goa	Clothing	Kids Wear	Kids Wear Repellat	1	7676

5000 rows × 15 columns

In [30]:
```
df["Profit_Margin"]=(df["Profit"] / df["Sales"]) * 100
```

In [31]:
```
df
```

Out[31]:

	Order ID	Order Date	Customer Name	Region	City	Category	Sub-Category	Product Name	Quantity	Un Pric
0	10001	2024-10-19	Kashvi Varty	South	Bangalore	Books	Non-Fiction	Non-Fiction Ipsum	2	3629
1	10002	2025-08-30	Advik Desai	North	Delhi	Groceries	Rice	Rice Nemo	1	4216
2	10003	2023-11-04	Rhea Kalla	East	Patna	Kitchen	Juicer	Juicer Odio	4	6487
3	10004	2025-05-23	Anika Sen	East	Kolkata	Groceries	Oil	Oil Doloribus	5	3732
4	10005	2025-01-19	Akarsh Kaul	West	Pune	Clothing	Kids Wear	Kids Wear Quo	1	5003
...	...	...	...	...	...	...	...	...	...	
4995	14996	2024-06-25	Nishith Kulkarni	East	Kolkata	Books	Fiction	Fiction Veritatis	3	6067
4996	14997	2024-12-22	Aaina Chander	North	Jaipur	Toys	Doll	Doll Nulla	5	7004
4997	14998	2025-04-15	Dhanush Gara	South	Bangalore	Beauty	Lipstick	Lipstick Eaque	1	4216
4998	14999	2024-07-08	Divyansh Malhotra	East	Kolkata	Electronics	Smartwatch	Smartwatch Adipisci	4	1356
4999	15000	2024-02-04	Aarush Walla	West	Goa	Clothing	Kids Wear	Kids Wear Repellat	1	7676

5000 rows × 16 columns

In [34]:
```
df["Order_Month"]=df["Order Date"].dt.month
```

In [35]:
```
df
```

Out[35]:

	Order ID	Order Date	Customer Name	Region	City	Category	Sub-Category	Product Name	Quantity	Un Pric
0	10001	2024-10-19	Kashvi Varty	South	Bangalore	Books	Non-Fiction	Non-Fiction Ipsum	2	3629
1	10002	2025-08-30	Advik Desai	North	Delhi	Groceries	Rice	Rice Nemo	1	4216
2	10003	2023-11-04	Rhea Kalla	East	Patna	Kitchen	Juicer	Juicer Odio	4	6487
3	10004	2025-05-23	Anika Sen	East	Kolkata	Groceries	Oil	Oil Doloribus	5	3732
4	10005	2025-01-19	Akarsh Kaul	West	Pune	Clothing	Kids Wear	Kids Wear Quo	1	5003
...	...	...	...	...	...	...	...	...	...	
4995	14996	2024-06-25	Nishith Kulkarni	East	Kolkata	Books	Fiction	Fiction Veritatis	3	6067
4996	14997	2024-12-22	Aaina Chander	North	Jaipur	Toys	Doll	Doll Nulla	5	7004
4997	14998	2025-04-15	Dhanush Gara	South	Bangalore	Beauty	Lipstick	Lipstick Eaque	1	4216
4998	14999	2024-07-08	Divyansh Malhotra	East	Kolkata	Electronics	Smartwatch	Smartwatch Adipisci	4	1356
4999	15000	2024-02-04	Aarush Walla	West	Goa	Clothing	Kids Wear	Kids Wear Repellat	1	7676

5000 rows × 17 columns

In [ ]: